

CR-012: Dark Mode and Light Mode — Full System

1. Change Request Details

Item	Details
Change Request ID	CR-012
Title	Dark Mode and Light Mode for the Full System
Theme Type	Page / Section-Specific Theme
Priority	Medium
Difficulty	Medium
Change Type	Frontend Enhancement / UI Improvement
Affected System	Student Management System

2. Business Requirement

The Student Management System shall provide a Dark Mode / Light Mode button on every applicable page. However, the theme selection must be isolated by system section. A theme selected in the Login section shall not change the Admin or Student section. A theme selected in the Admin section shall not change the Student section, and vice versa.

3. Required Theme Behaviour

- Every applicable page shall display a Dark / Light theme toggle.
- Login theme is independent from Admin and Student themes.
- Admin theme is shared only across Admin pages.
- Student theme is shared only across Student pages.
- Admin Add Student and Edit Student pages shall use the Admin theme preference..
- Changing one section's theme shall not modify another section's theme.
- Theme preferences shall persist after page navigation and browser refresh.

4. Requested Change

- Add a Dark/Light Mode button to every applicable page.
- Keep Login, Admin, and Student themes separate.
- Admin pages follow the Admin theme; Student pages follow the Student theme.
- Save each theme preference using localStorage.
- Apply themes to all UI elements.
- No changes to authentication, authorization, student data, CRUD operations, APIs, or database functionality.

5. Scope

In Scope

- Dark/Light toggle on every applicable page.
- Independent Login theme.
- Independent Admin theme.
- Independent Student theme.
- Admin child pages inherit the Admin theme.
- Student child pages inherit the Student theme.
- Theme persistence using localStorage.
- Reusable ThemeToggle component.
- Readable backgrounds, text, cards, forms, tables, buttons, navigation and modals in both themes.
- No interruption to existing system functionality.

Out of Scope

- Backend changes.
- Database schema changes.
- New theme-related APIs.
- Authentication or authorization changes.
- Student data changes.
- Changes to existing CRUD business logic.

6. Example Page Behaviour

Page	Theme Setting	Affected Area
Login	Login Theme	Login section only
Admin Dashboard	Admin Theme	All Admin pages
Admin Add Student	Admin Theme	Admin section
Admin Edit Student	Admin Theme	Admin section
Admin Student List	Admin Theme	Admin section
Student Dashboard	Student Theme	All Student pages
Student Profile	Student Theme	Student section
Student Results	Student Theme	Student section
Student Attendance	Student Theme	Student section

7. Expected / Potentially Affected Files

File / Location	Purpose
frontend/src/main.jsx	Application setup; preserve existing routing and providers.
frontend/src/context/ThemeContext.jsx	Manage section-specific theme state.
frontend/src/components/ThemeToggle.jsx	Reusable Dark / Light button.
frontend/src/pages/LandingPage.jsx	Theme control if included in the Home section.
frontend/src/pages/LoginPage.jsx	Login-specific theme.
frontend/src/pages/DashboardPage.jsx	Admin-specific theme.

frontend/src/pages/StudentDashboardPage.jsx	Student-specific theme.
frontend/src/components/Navbar.jsx	Display the relevant theme toggle where applicable.
frontend/src/styles/global.css	Theme variables and common styles.
frontend/src/styles/landing.css	Home theme styling.
frontend/src/styles/login.css	Login theme styling.
frontend/src/styles/dashboard.css	Admin theme styling.
frontend/src/styles/studentDashboard.css	Student theme styling.

8. Functional Requirements

1. FR-001: Every applicable page shall contain a Dark/Light theme toggle.
2. FR-002: Login theme shall be stored separately from Admin and Student themes.
3. FR-003: Admin theme shall apply to all Admin pages.
4. FR-004: Student theme shall apply to all Student pages.
5. FR-005: Admin theme changes shall not change Student or Login themes.
6. FR-006: Student theme changes shall not change Admin or Login themes.
7. FR-007: Login theme changes shall not change Admin or Student themes.
8. FR-008: Theme switching shall occur without a page reload.
9. FR-009: Theme preferences shall persist after navigation.
10. FR-010: Theme preferences shall persist after browser refresh.
11. FR-011: Existing authentication, CRUD, search, forms and data operations shall remain unchanged.

9. Security Requirements

- Theme preference shall not contain passwords, tokens or sensitive information.
- Authentication and authorization logic shall remain unchanged.
- Admin access controls shall remain unchanged.
- Student access controls shall remain unchanged.
- No new database or API endpoint is required for theme preferences.

10. Acceptance Criteria

12. AC-001: Dark/Light button is visible on every required page.
13. AC-002: Login theme can be changed independently.
14. AC-003: Admin theme can be changed independently.
15. AC-004: Student theme can be changed independently.
16. AC-005: Admin child pages use the selected Admin theme.
17. AC-006: Student child pages use the selected Student theme.
18. AC-007: Changing Admin theme does not change Student theme.
19. AC-008: Changing Student theme does not change Admin theme.
20. AC-009: Changing Login theme does not change Admin or Student theme.
21. AC-010: Each section remembers its own theme after refresh.
22. AC-011: Existing Login, Add, Edit, Delete and Search functions continue to work.

23. AC-012: No theme-related runtime errors are introduced.

11. Expected Outcome

The complete Student Management System will have a Dark/Light toggle on every required page while maintaining independent theme preferences for Login, Admin and Student sections. Admin pages will share the Admin preference, Student pages will share the Student preference, and Login will maintain its own preference. A change in one section will not unexpectedly affect another section.

12. Implementation Report Requirement

- Implementation summary.
 - New and modified files.
 - Section-specific theme architecture.
 - ThemeToggle implementation.
 - localStorage keys and persistence behaviour.
 - CSS variables and Dark/Light styling.
 - Testing results for Login, Admin and Student sections.
 - Screenshots of each section in both themes.
 - Acceptance criteria verification.
 - Final status and outstanding issues.
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